



Exercise 2

Optimizations

Please answer the following questions based on the Video we just watched together.

- a) What makes the XGBoost tree construction algorithm “greedy”?
- b) What is the difference between the “approximate greedy algorithm” and the “exact greedy algorithm”?
- c) For Regression with XGBoost, what is the difference between weighted quantiles and normal quantiles?
- d) For Classification with XGBoost, what is the difference between weighted quantiles and normal quantiles?
- e) What is the difference between the “cover” he mentions and the “weights”?

- a) immediately evaluates split (local best) and does not care for possible future improvement (global best)
- b) approx looks at quantiles instead of all possible split points, exact does the opposite
- c) none bc weight is always 1 for all values
- d) weighted quantiles split groups based on sum of weights instead of num of observations
low-confidence values get a higher weight than high confidence ones
- e) none, they are the same and he just calls them differently